2026–2027 AIDaS Seed Grant Program

Semi-Autonomous AI Research on the Wayne State Grid: From Labeled CT to C-Arm Level Localization, with Medical Trainees Driving the Agents

Strategic areas	Foundational AI (area 1, primary; Aim 1): the reliability of agentic systems, studied in agents doing research held to sealed tests, checked code and reproducible records. AI for Education and Workforce Training (area 3; Aim 2): medical trainees doing computational research, with literature answers grounded in verbatim, cited quotations. AI for Health (area 4; Aim 3): intraoperative imaging decision support, an advisory level label per C-arm image that the surgeon confirms and the stack withholds when its checks disagree.
Principal Investigator	Zhizhong Han, PhD, Assistant Professor, Department of Computer Science, College of Engineering, Wayne State University; founding director of the Machine Perception Lab; AIDaS affiliated member. Contact: [DATA NEEDED: office, phone and e-mail].
Co-Principal Investigator	Loren Schwiebert, PhD, Professor, Department of Computer Science, College of Engineering, Wayne State University; AIDaS affiliated member. Contact: [DATA NEEDED: office, phone and e-mail].
Clinical collaborator	Gregory Schwing, MD, PhD, General Surgery, Wayne State University School of Medicine / Detroit Medical Center; leads the Open Spine Consortium. Receives no award funds.
New Investigator status	The PI is a New Investigator: an Assistant Professor at Wayne State since January 2021 [DATA NEEDED: confirm that the appointment is tenure-track], and an NIH Early Stage Investigator until November 2027, ten years after the PI's PhD of November 2017 [DATA NEEDED: confirm the end date in eRA Commons]. The Co-PI is a Professor and not a New Investigator.
Request	$18,568 for 12 months, January 1 to December 31, 2027. No award funds go to an external collaborator: lines B2, B5, C1 and D buy goods and services, and Aim 3 uses WSU's own C-arm. The PI leads no other AIDaS proposal this cycle [DATA NEEDED: confirm].

Abstract. Wrong-level spine surgery persists because surgeons confirm the level by counting vertebrae on an X-ray from a C-arm, the operating room's mobile fluoroscope. The count fails in patients with an extra or missing lumbar segment and on images that show too few vertebrae. The Open Spine Consortium at Wayne State has built the parts of an automated check: 802 CTs with every vertebra labeled, a renderer that turns them into simulated C-arm images, and a level model. Never trained on a C-arm image, it names the level of 96.4 % of the lateral vertebrae a film-trained detector finds on a public C-arm dataset (88.0 % counting those it misses). Detection is the gap this project attacks first: a second detector, trained on two other devices' frames, finds 80.5 % of the vertebrae on that dataset's C-arm (90.6 % at a lower confidence threshold), against the 97 % set in advance. Closing the gap takes hundreds of GPU experiments on a shared computing cluster, more than one person can run. Aim 1 has an AI agent iterate the scientific method inside the consortium's frameworks (the cluster's job rules, a gate on its code, reproducibility and reporting protocols, coordination of several agents), and a result counts only when a test sealed before the run passes and a person merges it. On tasks with a planted fault, we expect it to submit a wrong result in under a fifth of runs, against over half without the frameworks' checks. Aim 2 gives the agent to eight medical trainees and measures, task against matched task, what it adds over a coding assistant, the 2026 program's setup. Their literature questions go to a language model on the grid, which quotes Scopus abstracts verbatim, with citation and DOI, under Elsevier's text and data mining terms, and no passage reaches an outside AI service. Aim 3 turns the agent on the detection gap, with learned X-ray simulation, and tests the result on CT-paired cadaver images taken at WSU. For the clinical validation after the award, an IRB protocol awaiting submission would add about 4,000 Detroit Medical Center spine surgery patients' CTs, operative notes and radiology reports, with any transitional vertebra recorded. Deliverables: Aim 1's tasks, records and report (MIT license, as an invention-disclosure review clears), the cadaver set, an FDA Pre-Submission, and NSF CAREER and NIH R21 proposals.

Keywords. agentic systems; AI for science; reproducibility; high-performance computing; medical education; fluoroscopy; 3-D Gaussian splatting; domain transfer; spine surgery

Project narrative

1. Significance and innovation

Surgeons confirm the level of a spine operation by counting vertebrae on an X-ray from a C-arm, the operating room's mobile fluoroscope. The count fails often enough to matter: a 27-hospital Japanese registry recorded wrong-level surgery in 0.12 % of 34,544 patients having degenerative spine surgery (0.42 % in the thoracic spine) [1], and about half of responding US neurosurgeons report at least one wrong-level operation [2,3]. It fails most on a transitional vertebra, part lumbar and part sacral: 12.9 % of 550 surgical patients had one, and their levels were miscounted in clinic in 54 % of them [4]. The best-studied aid, LevelCheck, registers the patient's preoperative CT to each image [5] in 17 to 72 s in an operating room [6], inside or at the edge of the 20 s to one minute surgeons said they would wait [7].

The Open Spine Consortium at Wayne State has built the parts of a check that needs no CT, only the C-arm image: CTSpinoPelvic1K, an open dataset of 802 CTs with every vertebra and the sacrum labeled [8]; a physics renderer that turns them into C-arm images; and a level model (Figure 3a). With no C-arm image in training, it names the level of 96.4 % of the lateral vertebrae a film-trained detector finds on a public C-arm dataset (Figure 4a). Detection does not yet carry to a new C-arm: on one held out of its training, our C-arm detector finds 80.5 % of vertebrae (90.6 % at a lower threshold), short of the 97 % set in advance (Figure 4b). Closing that gap takes hundreds of GPU jobs run, checked and recorded on a shared cluster (one results figure took 250 fine-tunes). The consortium's 2026 trainee program hit the same wall: it relied on mentors expert in Python, imaging and cluster computing [9].

Our premise is that the bottleneck has moved from the model to the experiments, and that an AI agent can run them if strict frameworks, not the model, decide which results count. Three aims test it (Figure 1). **Aim 1** has the agent iterate the scientific method inside the consortium's frameworks. Beethoven runs Claude Code on the grid and keeps the grid's profile as checked data and its job rules in writing; beethoven-qa checks the agent's code; beethoven-research holds REFORMS [10] rules for machine-learning science and plans the CLAIM [11] checklist for imaging AI; and Symphony gives each piece of work to one agent. The methodological contribution measures *framework-bound scientific iteration*: how many wrong but plausible results a research agent submits when the rules are only written, and when checks enforce them. Benchmarks score agents' finished code edits [12], machine-learning engineering [13] or papers [14], and studies of research agents probe their leakage and tampering [15,16]; we plant the faults an agent inherits from a lab's data and code, and count those it passes on. **Aim 2** puts the agent in medical trainees' hands through Beethoven (Figure 2). **Aim 3** turns it on the detection gap, with a paired CT and C-arm cadaver set imaged at WSU and a differentiable 3-D Gaussian X-ray renderer, as in [17,18], built on the PI's Gaussian-splatting work.

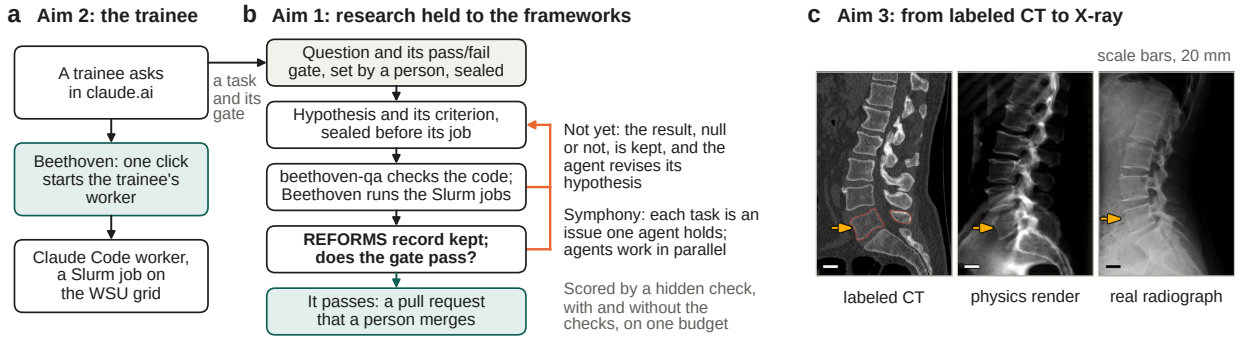


Figure 1. One loop, three aims. (a) Beethoven starts the trainee's agent. (b) The agent iterates hypotheses inside the frameworks until a sealed gate passes or its budget runs out. (c) Why counting fails: a sixth lumbar vertebra (arrow) in a labeled CT, its render and another person's radiograph.

2. Aims, approach and preliminary work

Aim 1 (Foundational AI): semi-autonomous research under strict frameworks. The agent is Claude Code at a pinned model version. Semi-autonomous: a person sets each task's question and gate (a pass/fail test, sealed by hash before the first job, that the agent can read but not change) and merges each result; any other step a person takes, blind to faults, is a logged intervention. The agent iterates: it seals each hypothesis with its criterion, runs jobs under Beethoven's rules, counts a result only from code beethoven-qa passed (contracts searched for counterexamples by CrossHair and Z3) and keeps its REFORMS record, until the gate passes or the budget is spent, then marks its Symphony pull request ready. *Approach.* Thirty tasks, 10 for development, come from questions Aim 3 and the consortium have answered. Each of the 20 sealed has a hidden check, a known answer the agent cannot read, and a clean and a faulted copy: once the checks freeze, someone who did not write them plants a fault, a patient in both training and test sets, the wrong label scheme, a threshold tuned on the test set, a typed number, or a run that does not reproduce. Both arms keep the model, Beethoven, the written rules and eight GPU-hours a task, reruns included; one adds the checks (sealed hypotheses, beethoven-qa, split and rerun checks); each runs each copy once, in its own repository (80 runs). *Preliminary work.* Built: Beethoven's grid profile and its Claude Code worker (a Slurm scheduler

job), beethoven-qa's gate on Beethoven's agent and research code, the patient-split check, and Symphony (public, MIT), whose tests catch all 93 mutated rules. The consortium builds the other REFORMS checks, the CLAIM checklist and one-command reruns of gated code in kind. *H1:* the checks cut wrong results on faulted copies from over half to under 20 %, with no fewer passes on clean copies, and each accepted number reruns within a sealed tolerance.

Aim 2 (Education and Workforce): medical trainees drive the agent. *Approach.* Eight medical students and residents in two waves of four [DATA NEEDED: confirm] each receive a grid account, Beethoven and a gated sub-project, their own or from Aim 3's list. The agent runs the experiments inside the frameworks; the trainee owns the question and the result, which a mentor reviews. Literature questions go to a language model on the grid that quotes verbatim, with citation and DOI, from indexed spine-related Scopus abstracts, under the text and data mining and AI rights Elsevier approved for WSU; no passage trains a model or reaches an outside AI service, Claude included; a misquoted passage fails a gate, and the mentor marks any quotation that does not support the answer. In week 1 each trainee does two matched starter tasks, none of Aim 1's: one with the agent, one with the 2026 setup (Claude as a coding assistant, the trainee submitting the jobs), task and order counterbalanced. The 2026 comfort survey runs before and after each wave; an IRB determination is requested in month 1. *Preliminary work* [9]: in 2026, 16 students, seven with no coding and eleven with no imaging experience, worked on ten sub-projects in that setup, with mentors expert in Slurm. All six domains improved on retrospective before-and-after ratings (Figure 2c), and all ten sub-projects went to a national meeting as abstracts. The report names what it lacked: a control group and objective measures. Aim 2 adds both. *H2:* with the agent, trainees pass more starter gates without a mentor's edit and reach each task's first passing gate sooner; with eight trainees an estimate with 95 % intervals that sizes the CyberTraining proposal.

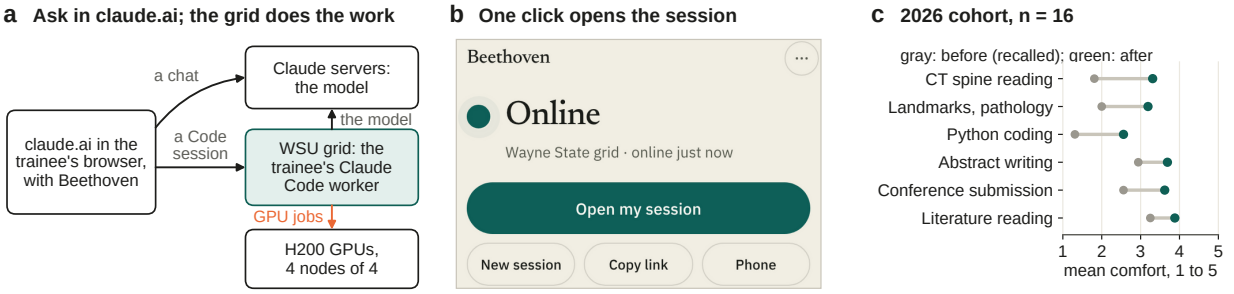


Figure 2. Beethoven, the trainee's door to the grid. (a) The trainee asks in claude.ai; the Claude Code worker runs as a Slurm job on the WSU grid, built to send GPU jobs to H200s. (b) One click opens the session. (c) The 2026 cohort's mean comfort, before (recalled) and after (n = 16) [9].

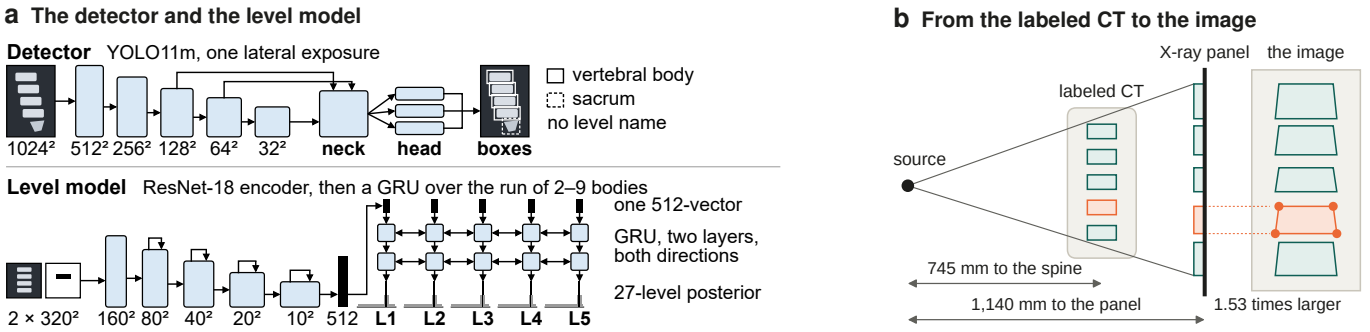


Figure 3. The level model, and how a CT labels the image. (a) The C-arm detector boxes the vertebral bodies and the sacrum, and the level model names each body among 27 levels. (b) Each labeled body and its corners project to the image; a transitional one in orange.

Aim 3 (AI for Health): labeled 3-D structure into 2-D fluoroscopy, at low latency. *Data.* Physics renders of CTSpinoPelvic1K, VerSe and TotalSegmentator CTs (7,045 standard and 1,482 transitional annotated frames) and a device-styled corpus of 142 TotalSegmentator spines; real frames of three devices: C-ArmLSpine (149 public lumbar cases, each a cone-beam CT with AP (front-to-back) and lateral frames, truth automatic: two CT labelers agreeing), DeepFluoro (cadaver pelvis fluoroscopy) and COBRA (Elekta radiotherapy imaging). *Model* (Figure 3a): a detector finds each body and the sacrum, a level model names each body, a count up from the sacrum checks the names, and the stack abstains where count and shape disagree or the name is unsure. New here, the C-arm detector will learn each body's four corners, as the film detector does, from the CT's corners projected at the device's geometry (Figure 3b), exact on renders. *Preliminary results* (Figure 4). With no C-arm image in training, the level model names 96.4 % of the 617 lateral bodies the film-trained detector finds on C-ArmLSpine (88.0 % of all 676 with agreed truth) but 43.7 % of the 631 AP bodies. Fine-tuned on the AP frames of five cases, it names 90.8 %, and 98.1 % with all other cases, each test case held out. On the 233 frames where the sacrum is found (81.5 % of the 286 that show one), cross-validated models, checked against the count, name 98.2 % of 1,009 bodies, and the stack abstains on 9.9 % of those frames, catching five of the six it would have named wrongly. Detection does not carry: trained on two

other devices' real frames, the C-arm detector finds 80.5 % of C-ArmLSpine's vertebrae and 76.8 % of its sacra at 0.74, a cross-device threshold chosen on other devices (90.6 % and 83.9 % at 0.25); on truth rebuilt after that scoring it finds 82.2 % of the 287 and 93.9 % of the 214 whose truth is complete and at least 80 % visible. *Closing the gap.* Today's renders hurt: added to real frames, styled renders lower detection from 80.5 to 63.1 % at 0.74 (90.6 to 87.8 % at 0.25), and a classifier tells them from real frames 99 to 100 % of the time. Physics simulation (DeepDRR [19]) with strong domain randomization has matched real-data training for hip landmarks in cadaver X-rays (SyntheX [20]). We randomize devices' geometry and spectra likewise and hold out each device in turn. We also fit the Gaussian renderer to a subset of the labeled CTs, learning only spectrum, scatter and sensor response until that classifier is at chance, so the anatomy and its labels stay exact. *Cadaver set.* Two specimens from WSU's Body Bequest Program are CT-scanned and imaged AP and lateral at lumbar and thoracic positions on WSU's C-arm. A frame grabber records the video output, the set is sealed by hash before any model sees it, and registering each specimen's CT to its frames [5] projects the CT's labels onto every frame. The intended use is an advisory level label on each C-arm image, which the surgeon confirms. End-to-end latency is unmeasured; Aim 3 measures it first. *H3:* on each held-out C-arm the detector finds 97 % of vertebrae and of sacra with no more false detections than today's at 0.74; the stack names a frame in under 1 s on a workstation GPU; and on every cadaver frame it names the right level or abstains.

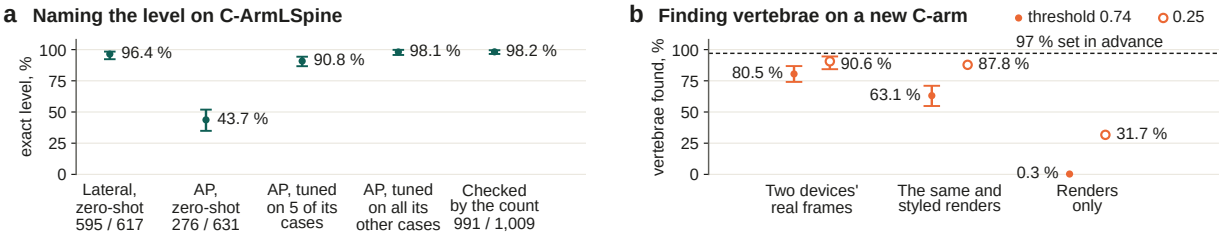


Figure 4. Held-out results; bars are 95 % case-clustered intervals, where recorded. (a) Level named exactly: zero-shot points never saw C-ArmLSpine, the others hold out whole cases. (b) Vertebrae found, of 287, on a C-arm never trained on, at 0.74 (filled) and 0.25 (open).

3. Evaluation: data, metrics and design

Aim	Data	Metrics	Design and analysis
1	30 answered tasks (10 for development); 20 sealed, each with a hidden check and a clean and a faulted copy	wrong results submitted; clean copies passed; faults caught, by kind and check; numbers that rerun; interventions; GPU-hours per accepted result; rule violations and gaming	with and without the checks, one model and budget; paired by task
2	8 trainees in two waves [DATA NEEDED: confirm]; two matched starter tasks each	starter gates passed without a mentor's edit; hours to each task's first passing gate; misquoted or unsupported quotations; comfort (secondary)	within-trainee crossover, task and order counterbalanced; plan preregistered
3	C-ArmLSpine, DeepFluoro and COBRA frames; the render banks; the sealed cadaver set	vertebra and sacrum recall at 0.74 and 0.25; false detections; exact level; abstention; corner error (mm); latency (median, 95th percentile); render-to-real separability	leave-one-device-out; case-clustered intervals; cadaver set opened once, models hashed first

H1 to H3 are written before any run; none is a powered test. H1 and H2 size the CAREER and CyberTraining proposals.

4. Expected outcomes, deliverables and timeline

Within the award: Aim 1's tasks, records and report; the cohort's outcomes; the cadaver set, deposited beside CTSpinoPelvic1K as the Body Bequest Program's terms allow, and a transfer and latency manuscript; an FDA Pre-Submission; and the CAREER and R21. The seed buys what does not exist yet: the researcher's hours, subscriptions, the cadaver set and regulatory review. The trainee program, grid, Beethoven, Symphony, pipelines and faculty time come in kind; the Gaussian renderer is fitted only as a feasibility test, its full build left to the R21. *Quarter 1:* missing checks built, tasks and gates sealed, faults planted, recruitment, bookings. *Quarter 2:* Aim 1's runs (months 4–8), wave 1 (months 4–7), cadaver imaging, transfer and latency results, Pre-Submission. *Quarter 3:* CAREER, wave 2 (months 8–11), the renderer's fit. *Quarter 4:* R21, final report, symposium. *Risks.* If the checks are not all ready by month 4, Aim 1 tests those built, wave 1 starts on the 2026 setup and the crossover moves to wave 2; agents run attended, a person approving commands blind to faults, until the grid clears unattended runs; a delayed specimen leaves the held-out devices as the test; a transfer result under 97 % becomes the R21's first aim.

References. [1] Furuya M, et al. Bone Joint J 2026;108-B(5):673-80. [2] Mody MG, et al. Spine 2008;33(2):194-8. [3] Groff MW, et al. World Neurosurg 2013;79(3-4):585-92. [4] Nakagawa T, et al. Tohoku J Exp Med 2017;242(3):223-8. [5] Otake Y, et al. Phys Med Biol 2012;57(17):5485-508. [6] Manbachi A, et al. Ann Biomed Eng 2018;46(10):1548-57. [7] De Silva T, et al. Spine 2016;41(20):E1249-56. [8] Open Spine Consortium. CTSpinoPelvic1K, release 11. doi:10.5281/zenodo.22797014. [9] OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research. Cureus 2026;18(7):e112661. [10] Kapoor S, et al. Sci Adv 2024;10(18):eadk3452. [11] Tejani AS, et al. Radiol Artif Intell 2024;6(4):e240300. [12] Jimenez CE, et al. SWE-bench. ICLR 2024. [13] Chan JS, et al. MLE-bench. ICLR 2025. [14] Lu C, et al. The AI Scientist. arXiv:2408.06292. [15] Luo Z, et al. arXiv:2509.08713. [16] Atinafu Y, Cohen R. RewardHackingAgents. ICDE Workshops 2026. [17] Cai Y, et al. X-Gaussian. ECCV 2024. [18] Zha R, et al. R^2-Gaussian. NeurIPS 2024. [19] Unberath M, et al. DeepDRR. MICCAI 2018. [20] Gao C, et al. Nat Mach Intell 2023;5:294-308.

AIDaS funding priority alignment

Target agencies, programs and solicitations

1. **NSF CAREER** (NSF 22-586; CISE, Division of Information and Intelligent Systems), PI Han, built on Aims 1 and 2: framework-bound agents as the research plan, the trainees as the education plan. Aim 1's agent then runs the 3-D-to-2-D experiments the PI leads in Aim 3, so it serves the PI's own vision program. Deadline July 28, 2027; eligibility is judged at the deadline (an untenured Assistant Professor with no earlier CAREER award) [DATA NEEDED: the PI's tenure timing and earlier CAREER competitions]. If the PI is not eligible, the same plan goes to IIS's core research under CISE's Future CoRe solicitation (NSF 25-543).
2. **NIH R21 at NIBIB**, built on Aim 3: the Trailblazer Award for New and Early Stage Investigators if NIBIB reissues it (forecast FOR-EB-26-001), or else NIBIB's Exploratory/Developmental Bioengineering Research Grant (PAR-25-321) or the parent R21 (PA-25-304), both open through October 2027. Submitted October 18, 2027 (October 16 is a Saturday), while the PI is still an Early Stage Investigator. Preliminary data: the cadaver set, the held-out-device result and the measured latency; its clinical validation, shaped by FDA's Pre-Submission feedback, draws on the consortium's pending protocol IRB-26-03-8648 (about 4,000 adult DMC spine surgery patients, their CT and records), where 99 patients give 80 % power to show a Wilson lower bound of 90 % if the true accuracy is 97 %.
3. **NSF CyberTraining** (NSF 23-520), a Pilot of up to $300,000 over two years led by Co-PI Schwiebert: agent-assisted training on advanced computing, sized by Aim 2's estimates. Deadline January 20, 2028; submitted in any case.

Relevance to national science and technology priorities

America's AI Action Plan (July 2025) asks for investment in AI-enabled science and, in its worker-first agenda, for AI skills in the workforce, which Aim 2 builds in Michigan's medical trainees. Executive Order 14363 (November 2025) launches the Genesis Mission, which joins AI agents to national computing, and NSF's Dear Colleague Letter 26-023 invites proposals to existing programs that support it. NIH's Bio Genesis Mission carries it into biomedicine. Aim 1 is that agenda at a university's scale: an agent running science on shared computing, held to verifiable gates. Executive Order 14303, Restoring Gold Standard Science (May 2025), and OSTP's guidance under it (June 2025) ask agencies to promote pre-registration of study protocols and transparent reporting of null results; every gate in this project is pre-registered by hash, and failed gates are reported. NIH's data sharing policy and its Strategic Plan for Data Science 2025–2030 ask for data others can reuse, as CTSpinoPelvic1K is and the cadaver set will be as its terms allow.

Team, expertise and roles

The PI directs the project, writes the FDA Pre-Submission, leads the CAREER and R21 proposals, designs Aim 1 and builds the 3-D-to-2-D model, including the Gaussian X-ray renderer; a computer vision researcher, the PI leads the Machine Perception Lab and publishes on 3-D Gaussian splatting and neural implicit surfaces (CVPR 2025 and 2026, NeurIPS 2024). The PI's part in the consortium's work so far: [DATA NEEDED: the PI's role, and since when]. The Co-PI, a high-performance computing researcher who accelerates scientific codes on GPUs, builds the evaluation harness, plants the faults and keeps the grid measures, and has four such papers with the clinical collaborator (2021–2024). The clinical collaborator leads the Open Spine Consortium, whose dataset, models, Beethoven, Symphony and trainee program [9] this proposal builds on, builds the missing checks in kind, and runs the cohort, the IRB determination, the imaging session, the literature index and the pending protocol. Beethoven's repositories are the clinical collaborator's proprietary software, given in kind; no award funds go to them. Line A's researcher checks Aim 1's runs and curates the cadaver set. Effort: PI [DATA NEEDED: %], Co-PI [DATA NEEDED: %], in kind.

Timeline for external submission

When	Milestone
January–March 2027 (months 1–3)	One-page concepts to the NSF IIS and NIBIB program officers; Aim 1's tasks and gates sealed; the Pre-Submission package written and reviewed
April 2027 (month 4)	FDA Pre-Submission filed; written feedback due within 70 days
July 28, 2027 (month 7)	NSF CAREER submitted, with Aim 1's early results and Aim 2's first wave
September 2027 (month 9)	The sealed cadaver set opened, once, for the R21
October 18, 2027 (month 10)	NIH R21 submitted, with the cadaver set, the latency and FDA's feedback
December 2027 (month 12)	Project ends; AIDaS final report and symposium
January 20, 2028 (month 13)	NSF CyberTraining Pilot submitted, Co-PI leading
2028	R21 resubmission at the first standard date after its summary statement; CAREER resubmission July 26, 2028, if eligible; NSF Smart Health when its next solicitation opens

The CAREER and R21 meet the expectation of a federal proposal within 12 months of the project's end, and the CyberTraining Pilot follows whatever their outcome; publications and proposals from the award acknowledge AIDaS.

Budget and justification

Line	Item	Basis	Amount
A	Part-time researcher, hourly	520 h at $15.00: Aim 1 180 h, Aim 2 160 h, Aim 3 180 h	$7,800
A2	Fringe on line A	FICA, 7.65 %	$597
B1	Two cadaver specimens, Body Bequest Program	placeholder $1,250 each [DATA NEEDED: program fee]	$2,500
B2	Specimen CT, two scans	placeholder $300 each [DATA NEEDED: facility and rate]	$600
B3	C-arm session, 0312 Elliman, through DLAR	4 h at a placeholder $300 [DATA NEEDED: DLAR rate; that the room takes human specimens]	$1,200
B4	Dosimetry and OEHS review	placeholder [DATA NEEDED: OEHS fee]	$300
B5	Frame grabber for the C-arm's video output	Magewell USB Capture AIO, retail price	$619
C1	Claude Team Premium seats, in their users' names	6 seats billed annually at $180 each	$1,080
C2	Data management and storage	2 TB of C&IT storage at $400 per TB-year; Zenodo free	$800
D	FDA Pre-Submission: regulatory consultant	10 h at $300; FDA charges no fee	$3,000
F	AIDaS symposium poster	36 by 48 in at $6 per sq ft	$72
	Total requested		**$18,568**

- **A, A2.** An hourly graduate student assistant (the call's graduate student support), or else a post-baccalaureate researcher: 520 hours, most in months 4–9 at up to 20 a week, at $15.00, Michigan's minimum wage from January 1, 2027 (MCL 408.934) [DATA NEEDED: WSU's job category and scale]. Temporary employees and students are not benefit-eligible, so A2 is FICA alone: 7.65 % for a hire who is not an enrolled student; hiring a student, at WSU's composite rate of 1.7 %, would buy about 30 more hours. Aim 1's hours check 80 runs against their hidden checks and records, two hours each, and seal the known answers in 20 hours; Aim 2's go to the cohort's accounts and measures; Aim 3's are 100 curating the cadaver set (de-identifying, labeling, sealing by hash) and 80 checking the agent's leave-one-device-out and latency runs against their gates. At $18 an hour the total would be $20,247, over the cap unless the hours fall to 507.
- **B1 to B5.** No public dataset we know of pairs a diagnostic CT of the thoracolumbar spine with C-arm frames of the same body: C-ArmLSpine pairs lumbar cone-beam CTs with frames posed by edge registration, and DeepFluoro pairs cadaver CTs with fluoroscopy of the pelvis. The cadaver set adds WSU's own C-arm. The Body Bequest Program's donor authorization permits de-identified images [DATA NEEDED: the program's terms for public release]. Each C-arm user enrolls in WSU's dosimetry program and files the training form of OEHS, WSU's safety office, both with the director of DLAR 15 business days before the session. The frame grabber records what the C-arm shows, the input a deployed tool would read [DATA NEEDED: the C-arm's model and video output]; at $619 it is far under WSU's $5,000 capital threshold and becomes WSU property.
- **C1, C2.** Anthropic's Team plan for scientists lets a verified PI add Premium seats at $15 a month, $180 a year billed annually: four for the trainees, reassigned in wave 2, and two for the people who start Aim 1's runs; the frameworks use no API key. All six are bought in January 2027, since the price holds 12 months from sign-up [DATA NEEDED: P-card route and the PI's verification]. The WSU grid's compute is free to WSU researchers and given in kind. Storage holds the render banks, the cadaver set and Aim 1's run records for the award year; what review clears goes to Zenodo at its end.
- **D.** FDA charges no fee for a Pre-Submission filed before October 1, 2027, when a fee proposed for fiscal year 2028 would begin. The filing asks FDA whether an advisory level label on each C-arm image, which the surgeon confirms, is the right intended use; whether the held-out-device and cadaver tests suffice as analytical validation; and what clinical study, against what reference standard, the R21 should plan. A mid-level consultant reviews the PI's package in months 1 to 3.
- **F.** Presenting at an AIDaS symposium is an award expectation; the poster is printed at Medical Communications.
- **Not requested.** PI and Co-PI salary, travel, tuition, capital equipment and indirect costs [DATA NEEDED: confirm AIDaS applies no F&A].
- **Allowability and dependencies.** The call's allowable costs include graduate student support, computing and data storage: lines A and A2 (with a graduate student assistant), C1 and C2, $10,277 in all, above the call's smallest award of $10,000. Lines B, D and F are research costs outside those headings [DATA NEEDED: the AIDaS office's answer]; B2 to B5 depend on B1. Without B the request is $13,349: Aim 3 tests on the held-out devices' frames, and its 100 cadaver hours go to Aim 1 repeats; without B and D, $10,349; without B, D and F, $10,277.

Biosketch: the Principal Investigator

The two-page NSF biosketch of the Principal Investigator goes here in the submitted PDF. Put the file below beside `main.tex` and compile again, and it replaces this page.

`biosketch_han.pdf`

Biosketch: the Co-Principal Investigator

The two-page NSF biosketch of the Co-Principal Investigator goes here in the submitted PDF. Put the file below beside `main.tex` and compile again, and it replaces this page.

`biosketch_schwiebert.pdf`